Physical Cume

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Enzyme kinetics is proposed to proceed by the following mechanism:

$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_2} E + P$$

where E is the enzyme, S is the substrate and P is the product.

(i) Transition-State theory shows that the rate constant is given by:

$$k = \frac{k_B T}{h} e^{-\Delta G^{o\ddagger}/RT}$$

where $\Delta G^{o\ddagger}$ is the standard molar Gibbs energy of activation energy, k_B is the Boltzmann constant, h is Planck's constant and T is the temperature. Using this equation, describe the thermodynamic arguments in favor of the above mechanism? **(25 pts)**

(ii) Prove that the initial rate, v_0 is given by:

$$v_0 = \frac{V_{Max}[S]}{K_M + [S]}$$

and also explain the meaning of V_{max} and K_M.

Clearly state any assumptions you make. **(50 pts)**

(iii) A carboxypeptidase enzyme has a K_M=2.0 μM and k_{cat}= k_2 =150 s^{-1}. Determine the concentration of the substrate which is needed to reach 80% of the maximum rate if the enzyme concentration is 10^{-8} M. **(25 pts)**